

lab_tour

DAY 2 · one continuous walking shot · quiet room

- **Say every location out loud as you reach it** — none of this reaches a caption otherwise
- Entry: **only enter when a supervisor is present**
- **Safety Corner — slow down.** East wall, near the NE corner, north side, by the biosafety cabinets
- There: shower · eye wash · extinguisher · spill kit · first aid · CHP binder · PPE · posted rules
- Eye wash: say **“fifteen minutes”** and hold the shot on it
- Benches · electrophoresis bench · thermocyclers · freezers, including the pink **Training1** box
- **Chemical hood** — flame sterilization, supervisors only · microwave · waste: white bin and benchtop beakers
- Exit route → **Mining Circle.** Walk outside and show it if you can
- **Do not pull the safety shower.** Point at it, say what it is for, move on

biosafety

DAY 2 · static, at a bench

- **Six lines that must be said, close to the page** — each maps onto a quiz question
- “Flush your eyes for **fifteen minutes**”
- “Evacuate to the **Mining Circle**”
- “**No food or drink anywhere in B144** — not even a capped water bottle”
- “**Never spray ethanol near an open flame**”
- “Prescription glasses count as eye protection”
- “**Supervisors Only** means do not touch it”
- Then: PPE head to toe · **BSL1 only** · the three signage tiers, and **scan a QR on camera** · report every spill · restore the bench

Examine and photograph plates

ctrl_read

DAY 2 · the overnight plates

- **Say every judgement out loud** — all of this is visual
- Name the sector that grew, and the ones that did not
- Sterility plates: contamination in the agar, bottle or hood
- Kanamycin — **rescued vs the no-rescue half**
- Cells-only plates should be empty
- Photograph · record batch date, who poured, result

parafilm

DAY 2 · the plates you are keeping

- **These plates are being held for picking later** — the seal is what makes that possible
- Cut a **1-inch strip along the long edge** of the roll
- Hold against the lid rim · stretch until it grips
- **Spin the plate**, keeping tension, to wrap the seam
- Press the end down onto itself to seal
- Inverted in the fridge · stable **1–2 weeks**
- **Get back to them within the week** — satellites keep growing and the green dims
- **Never parafilm anything going into the incubator** — cells need air

Pick BestP colonies into a block

bestp_pick

LATER · full 24-well block

- **Photograph the plates under blue light first** — the only record of what you picked from
- **4 mL 2YT + carb** per well
- **4 colonies per plate** · block is **4 × 6**
- **Left to right, in the sample order** — otherwise the data entry scrambles silently
- Airpore sheet on, then cover
- Multitron overnight
- Parafilm the plates · upside-down in the fridge

Pick pP6 clones into a 15 mL conical

pp6_pick

LATER · do 2

- Brightest green under blue light · **mark them on the plate**
- **Small and slow-growing can be the strongest** — do not just take the biggest
- Well-isolated only · watch for **satellites**, the white specks around a green colony
- **4 mL 2YT + carb**
- Sterile toothpick · **drop it in and leave it there**
- Shaking incubator overnight
- Parafilm the plate · upside-down in the fridge

Tecan the BestP block

bestp_tecan

DAY 3 · Tecan M Nano · 48 read wells

- Check the cultures grew and are fluorescent **before anything else**
- **100 µL** per culture into a **black-walled** 96-well plate
- P200 multichannel · **columns of 8** · 2 draws per sample
- Fluorescence: **Ex 483 / Em 525**, bandwidths 20, **gain 40**, 20 reads
- high sensitivity · 40 µs · Z 11019 µm · 30 °C · COS96ft
- **Also read OD600** — fluorescence alone is uninterpretable
- Save to the **USB stick**; that computer has no internet
- **Remove your plate before shutdown** · bleach and rinse it

What BestP is measuring

bestp_intro

WHENEVER · no bench work

- **RPU** = fluorescence per OD unit, normalised to **pJ01 = 1**
- **pJ12** very weak · **pJ01** the reference · **pJ19** very strong
- Same backbone, same reporter — so **only the promoter differs**
- Raw fluorescence is not comparable across runs; RPU is
- The goal: pP6 clones that **match or beat pJ19**